

METZ NANCY LORRAINE AP, France

WMO: 070930

Lat: 48.979N

Lon: 6.243E

Elev: 870

StdP: 14.24

Time Zone: 1.00 (EUC)

Period: 99-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.7	21.7	11.9	10.4	21.6	15.9	12.7	24.5	36.8	42.3	31.3	42.4	8.1	50	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	88.3	67.7	84.3	66.5	80.9	65.1	69.4	83.5	68.0	81.0	66.6	78.1	8.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.1	95.9	73.5	63.8	91.6	72.1	62.5	87.2	70.6	33.9	83.6	32.8	81.3	31.6	78.1	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
29.1	23.4	20.5	12.8	94.4	5.1	4.3	9.1	97.5	6.1	100.0	3.3	102.5	-0.4	105.6
			12.0	72.6	5.1	1.4	8.4	73.5	5.4	74.3	2.5	75.1	-1.2	76.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	51.2	35.6	37.5	43.6	50.3	57.0	63.4	67.0	66.3	59.6	52.1	43.1	37.5	(6)
(7)		DBStd	12.98	7.77	7.30	6.87	6.72	6.43	6.52	6.42	6.14	5.74	6.64	6.68	7.50	(7)
(8)		HDD50	1774	447	351	216	78	12	0	0	0	1	56	223	390	(8)
(9)		HDD65	5355	911	770	663	442	259	108	52	56	182	402	658	852	(9)
(10)		CDD50	2195	1	1	17	86	229	402	529	504	289	120	15	2	(10)
(11)		CDD65	299	0	0	0	0	11	59	115	94	19	1	0	0	(11)
(12)		CDH74	3009	0	0	0	16	126	599	1187	900	176	5	0	0	(12)
(13)		CDH80	975	0	0	0	0	20	177	424	325	29	0	0	0	(13)
(14)	Wind	WSAvg	9.7	11.1	11.5	11.1	9.4	8.6	8.0	8.6	8.2	8.6	9.7	10.1	11.2	(14)
(15)	Precipitation	PrecAvg	29.4	2.3	2.3	2.0	1.9	2.6	2.4	2.6	2.7	2.2	2.7	2.8	2.8	(15)
(16)		PrecMax	37.6	5.0	5.4	5.9	4.4	4.9	5.6	6.5	6.3	4.2	6.8	6.8	7.9	(16)
(17)		PrecMin	22.0	0.5	0.5	0.6	0.1	1.3	1.0	0.7	0.8	0.9	0.6	0.5	0.4	(17)
(18)		PrecStd	4.4	1.0	1.3	1.3	1.2	1.0	1.1	1.3	1.4	1.1	1.5	1.4	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.0	58.1	67.7	76.3	83.2	90.5	94.2	95.4	83.8	74.1	61.9	55.5	(19)
(20)		MCWB	50.9	48.9	52.6	59.7	67.2	68.8	68.3	68.7	65.5	61.7	56.4	51.5	(20)	
(21)		2%	DB	51.6	53.2	62.8	71.7	77.4	85.2	88.3	87.9	79.1	68.9	57.6	52.6	(21)
(22)		MCWB	49.3	47.3	50.8	57.1	62.7	67.6	67.1	68.1	64.3	60.4	53.7	49.8	(22)	
(23)		5%	DB	48.7	50.2	58.1	67.1	73.8	80.7	84.5	82.8	74.9	64.8	55.3	50.4	(23)
(24)		MCWB	46.5	46.3	49.4	55.1	61.1	65.7	66.3	66.0	62.6	58.7	52.2	48.1	(24)	
(25)		10%	DB	46.3	47.5	54.1	62.9	70.0	76.8	80.9	78.6	70.5	61.4	52.5	47.9	(25)
(26)		MCWB	44.5	44.5	48.1	53.1	59.3	64.2	64.8	64.5	60.6	56.8	50.0	46.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.1	50.6	55.1	61.7	68.6	72.0	70.7	71.0	67.4	63.8	57.4	52.7	(27)
(28)		MCDWB	53.7	55.0	63.5	73.3	80.2	84.9	85.8	87.4	79.1	70.2	60.2	54.3	(28)	
(29)		2%	WB	49.5	48.6	52.7	58.4	65.5	69.2	68.9	68.9	65.5	61.3	54.4	50.3	(29)
(30)		MCDWB	51.4	52.0	58.8	69.1	74.8	80.6	84.6	83.0	76.0	66.8	57.0	52.3	(30)	
(31)		5%	WB	46.8	46.7	50.7	55.9	62.5	67.4	67.5	67.4	63.7	59.3	52.4	48.2	(31)
(32)		MCDWB	48.5	49.6	56.3	65.0	70.6	78.1	81.7	79.8	72.5	64.2	55.0	50.0	(32)	
(33)		10%	WB	44.4	44.7	48.8	53.9	60.3	65.4	65.9	65.9	61.8	57.3	50.1	46.1	(33)
(34)		MCDWB	46.1	47.4	53.8	61.7	67.9	75.0	78.1	76.2	68.9	61.1	52.4	47.9	(34)	
(35)	Mean Daily Temperature Range		MDBR	8.8	11.3	15.4	18.2	18.7	20.0	19.4	18.8	17.7	14.0	9.7	8.4	(35)
(36)		5% DB	MCDBR	10.0	16.1	22.9	25.1	24.6	26.2	26.2	27.1	24.4	18.5	12.8	9.9	(36)
(37)		MCWBR	8.5	11.2	13.2	13.3	12.7	12.0	9.9	10.7	11.5	10.9	9.4	8.0	(37)	
(38)		5% WB	MCDBR	9.9	13.3	18.3	22.7	21.8	23.8	23.7	22.6	21.4	17.0	11.8	9.4	(38)
(39)		MCWBR	8.8	10.0	11.7	12.6	12.2	11.9	10.0	10.3	11.4	10.6	9.3	8.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.309	0.333	0.360	0.389	0.399	0.403	0.399	0.395	0.370	0.355	0.326	0.306	(40)	
(41)		taud	2.444	2.396	2.331	2.267	2.272	2.292	2.297	2.334	2.383	2.419	2.442	2.427	(41)	
(42)		Ebn at Noon	239	258	269	272	273	272	272	267	263	246	229	223	(42)	
(43)		Edh at Noon	20	27	33	39	41	40	39	36	31	26	20	18	(43)	
(44)	All-Sky Solar Radiation	RadAvg	275	512	897	1362	1610	1819	1714	1476	1124	658	316	220	(44)	
(45)		RadStd	43	91	134	178	177	159	169	154	120	55	33	37	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	-0.61	N/A	N/A	N/A	N/A	(46)
(47)	Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page